

1. Introduction

Machine Learning (ML) encompasses a suite of statistical and computational techniques that allow systems to learn patterns from data and make predictions without explicit programming. Broadly categorised into supervised and unsupervised learning, ML has found extensive applications in education, healthcare, business, and policy (Cortez & Silva, 2008). Supervised techniques — including regression and classification — require labelled data to train predictive models, whereas unsupervised techniques discover latent structure in unlabelled data through methods such as clustering and dimensionality reduction.

This report analyses the Student Performance Dataset (Cortez, 2014), which contains socio-economic, demographic, and academic information on secondary school students in Portugal. The dataset records performance in two subjects: Mathematics (395 students) and Portuguese (649 students). We combine both subjects (1,044 combined records) to increase sample size and test whether findings generalise across subjects. A total of 32 input features are available, including family background, lifestyle behaviours, and periodic academic grades (G1, G2), with the final grade (G3, scored 0-20) serving as the primary outcome of interest.

The report is structured around three analytical tasks. First, we apply Principal Component Analysis (PCA) and K-Means Clustering (unsupervised learning) to identify meaningful student profiles based on behavioural and demographic characteristics alone — without using grades. Second, we use regression to predict the continuous final grade G3, under two scenarios: including and excluding prior periodic grades G1 and G2. This dual-scenario design directly quantifies the information value of periodic assessments and evaluates the feasibility of enrolment-time prediction. Third, we use classification to predict whether a student will pass ($G3 \geq 10$) or fail, evaluating multiple models and analysing the optimal decision threshold for a school early-warning deployment context.

The following research questions (RQs) guide the analysis:

- RQ1 (Unsupervised): What are the optimal student behavioural clusters, and do they significantly predict academic outcomes?
- RQ2 (Regression): Which model best predicts final grade G3, and how much predictive information do prior grades G1/G2 contribute?
- RQ3 (Classification): Can we accurately identify at-risk students, and what threshold maximises detection of failing students?

2. Data Description and Exploratory Analysis

2.1 Dataset and Variables

The dataset (Cortez, 2014) is publicly available from the UCI Machine Learning Repository at <https://archive.uci.edu/dataset/320/student+performance>. It comprises two files: student-mat.csv (Mathematics) and student-por.csv (Portuguese Language). Each file contains 33 variables spanning four categories: student demographics (e.g., age, sex, address), family background (e.g., parental education Medu/Fedu, family size), lifestyle behaviours (e.g., study time, alcohol consumption Dalc/Walc, absences, going out frequency goout), and academic performance (G1, G2, G3). Both nominal and ordinal categorical variables are present alongside continuous numeric variables.

After label-encoding binary variables, one-hot encoding nominal variables (mother's job, father's job, reason, guardian), and converting grades to numeric format, the combined encoded dataset contains 1,044 rows and 42 columns with no missing values. Merging on shared demographic keys reveals that 382 students appear in both subjects; their cross-subject final grade correlation is $r = 0.48$,

suggesting that roughly half the variance in final grade is subject-specific rather than attributable to shared student characteristics alone.

2.2 Exploratory Data Analysis

A notable finding from the exploratory analysis is the large disparity in failure rates between subjects: Mathematics has a 32.9% failure rate ($G3 < 10$) compared to 15.4% in Portuguese — more than double. This motivates the combined analysis, as the Maths cohort provides a substantially harder prediction problem.

The correlation matrix over core numeric features reveals that the two periodic grades $G1$ ($r = 0.81$) and $G2$ ($r = 0.91$) are overwhelmingly the strongest predictors of $G3$, consistent with the literature (Cortez & Silva, 2008). Among non-grade variables, the most important correlates are: number of past failures ($r = -0.38$), mother's education level $Medu$ ($r = 0.20$), study time ($r = 0.16$), father's education $Fedu$ ($r = 0.16$), and workday alcohol consumption $Dalc$ ($r = -0.13$). Alcohol consumption (both daily and weekend) shows a consistent negative association with grades across the combined dataset.

Boxplot analysis confirms the monotonic relationship between study time and $G3$ (median grade rises with each study time category) and the steep decline in median grade with increasing past failures — students with 3 prior failures have a median final grade below the pass threshold. These two variables emerge as the primary actionable levers for targeted intervention.

Of the 1,044 combined records, 53 (5.1%) have $G3 = 0$. In the original dataset documentation, this value typically represents non-submission or voluntary withdrawal rather than a true grade of zero. We retain these records throughout the analysis, noting their potential influence on regression residuals.

3. Unsupervised Learning

3.1 Substantive Issue

Schools often lack structured tools for identifying student risk profiles at enrolment — before any academic grades are available. Clustering students into behavioural profiles using only demographic and lifestyle features (excluding grades) can reveal which students share characteristics predictive of poor performance, providing a basis for pre-emptive intervention allocation. This section addresses RQ1.

3.2 Methodology

Twenty-one behavioural and demographic features were selected, explicitly excluding $G1$, $G2$, and $G3$ so that clusters reflect student characteristics rather than outcomes: age, $Medu$, $Fedu$, traveltime, studytime, failures, famrel, freetime, goout, $Dalc$, $Walc$, health, absences, school, sex, address, internet, romantic, higher, activities, and paid.

All features were standardised (zero mean, unit variance) prior to analysis. Principal Component Analysis (PCA) was first applied to understand the latent structure of the feature space. K-Means Clustering was then applied with the optimal number of clusters selected using both the Elbow Method (within-cluster sum of squares) and the Silhouette Score. Cluster validity was assessed using the Kruskal-Wallis non-parametric test to determine whether clusters differ significantly in final grade $G3$ — a criterion not used in forming the clusters.

3.3 Principal Component Analysis

PCA was applied to the standardised 21-feature matrix. The scree plot reveals that 10 principal components are required to explain at least 90% of total variance — indicating a moderately high-dimensional feature space without strong redundancy. However, the first two components account for 31.9% of variance combined and provide useful low-dimensional visualisation.

Examining the loading vectors reveals a clear latent structure. PC1 loads most heavily (positively) on: Walc (weekend alcohol, loading = 0.37), Dalc (workday alcohol, 0.35), goout (going out, 0.32), and freetime (0.28), while loading negatively on studytime (-0.23) and internet (-0.18). This component can be interpreted as a 'risk behaviour' dimension: students with high PC1 scores engage heavily in social and alcohol-related activities at the expense of academic engagement. PC2 loads positively on Medu and Fedu (parental education), capturing a 'socio-economic background' dimension. This two-dimensional structure aligns with findings in the education literature: lifestyle factors and socio-economic background represent two orthogonal determinants of academic risk.

► PC1 represents a risk behaviour dimension (alcohol, socialising, low study time); PC2 represents socio-economic background. These two dimensions align with established theoretical frameworks in educational attainment research.

3.4 K-Means Clustering

Both the Elbow Method and Silhouette Score were applied over $K = 2$ to 10 clusters. The Elbow Method shows a marked reduction in within-cluster SSE from $K=1$ to $K=3$, with diminishing returns thereafter. The Silhouette Score is maximised at $K = 2$ (score = 0.118), indicating that a two-cluster solution best separates the data given the selected features.

The K-Means algorithm (10 random initialisations, random state fixed for reproducibility) was applied with $K = 2$. A Kruskal-Wallis H-test was conducted to validate whether the resulting clusters differ significantly in G3. The result ($H = 106.63$, $p < 0.0001$) confirms that the clusters are statistically associated with final grade, despite grades not being used in their formation — a key validation criterion.

Cluster	N	Pass Rate (%)	Mean G3	Failures (mean)	Walc (mean)	Medu (mean)	Studytime (mean)	Absences (mean)
0 — At-Risk	382	66.8%	9.93	0.55	2.57	1.78	1.69	4.00
1 — On-Track	662	84.4%	12.16	0.10	2.12	3.08	2.13	4.68

Table 1: K-Means Cluster Profiles ($K=2$)

The cluster profiles are substantively meaningful. Cluster 0 ($n=382$, 36.6% of students) is characterised by lower parental education (Medu = 1.78 vs 3.08), higher past failures (0.55 vs 0.10), higher weekend alcohol consumption (Walc = 2.57 vs 2.12), and lower study time (1.69 vs 2.13). Its pass rate of 66.8% is dramatically lower than Cluster 1's 84.4%, and its mean G3 (9.93) falls below the pass threshold. Cluster 1 ($n=662$) represents students with higher socio-economic backgrounds, better study habits, and lower-risk lifestyle profiles.

► The two clusters align with a socio-economically disadvantaged, high-risk-behaviour profile (Cluster 0) versus an educationally supported, low-risk profile (Cluster 1). The pass rate gap of 17.6 percentage points, confirmed significant by the Kruskal-Wallis test, validates these clusters as useful for identifying students who may benefit from targeted intervention at enrolment.

3.5 Results

RQ1: What are the optimal student behavioural clusters, and do they significantly predict academic outcomes?

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- The optimal number of clusters is $K = 2$ (Silhouette Score = 0.118), supported by the Elbow Method.
 - Cluster 0 represents an at-risk profile: lower parental education, more past failures, higher alcohol consumption, and lower study time. Mean G3 = 9.93, Pass Rate = 66.8%.
 - Cluster 1 represents an on-track profile: higher parental education, fewer failures, moderate alcohol consumption, and higher study time. Mean G3 = 12.16, Pass Rate = 84.4%.
 - The Kruskal-Wallis test ($H = 106.63$, $p < 0.0001$) confirms that clusters differ significantly in G3 despite grades not being used in their construction.
 - Policy implication: Cluster 0 membership at enrolment could trigger a support referral for monitoring, counselling, or study skills intervention before any grades are recorded.

4. Regression: Predicting Final Grade G3

4.1 Substantive Issue

Predicting a student's final grade enables schools to rank students by predicted performance and allocate resources proportionally. We address RQ2 by training regression models under two parallel scenarios: (A) including G1 and G2 as predictors — maximising accuracy as a benchmark — and (B) excluding G1 and G2 — simulating a model deployable at enrolment before any grades are available. Comparing Scenarios A and B directly quantifies the information value of periodic assessments and tests whether demographic and behavioural features alone provide actionable predictive power.

4.2 Methodology

Seven regression models were trained and evaluated under both scenarios using an 80/20 train-test split (random state fixed for reproducibility). Models included: Linear Regression, Ridge Regression ($\alpha=1$), Lasso Regression ($\alpha=0.01$), Decision Tree (max depth=6), Random Forest (200 trees), Gradient Boosting (200 estimators), and Support Vector Regression (RBF kernel, $C=10$, $\epsilon=0.5$). Linear and kernel-based models received standardised inputs; tree-based models used raw values. Performance was assessed using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), R-squared (R^2), and 5-fold cross-validated RMSE. The best model (Random Forest) was further tuned via 5-fold GridSearchCV over $n_estimators \in \{100, 200, 300\}$, $max_depth \in \{None, 5, 10\}$, and $min_samples_leaf \in \{1, 2, 4\}$.

Feature importance was assessed using permutation importance (15 repeats) rather than Gini impurity — the former is more reliable as it captures the actual impact of each feature on held-out test performance rather than training-set statistics.

4.3 Analysis

4.3.1 Scenario A — With G1 and G2

Including G1 and G2 substantially improves all models. The tuned Random Forest ($max_depth=5$, $min_samples_leaf=4$, $n_estimators=300$) achieves the lowest test RMSE of 1.669 with $R^2 = 0.820$, meaning 82.0% of variance in G3 is explained. Gradient Boosting achieves a marginally higher raw test RMSE (1.677) but slightly lower CV-RMSE (1.488), reflecting similar generalisation ability. Linear models (Lasso, Ridge, Linear) achieve respectable $R^2 \approx 0.79$ with $RMSE \approx 1.79$, demonstrating that much of the predictive signal from G1/G2 is linear.

4.3.2 Scenario B — Without G1 and G2

Removing G1 and G2 substantially degrades performance, as expected. The best model is SVR (RBF kernel), achieving $RMSE = 3.397$ and $R^2 = 0.254$. Random Forest achieves similar performance ($RMSE = 3.702$, $R^2 = 0.114$). Notably, linear models fail entirely ($R^2 \approx 0.09$), suggesting that the remaining relationship between demographic/behavioural features and G3 is largely non-linear. The SVR's superior performance in this scenario reflects its ability to capture complex non-linear relationships through the RBF kernel.

Model	RMSE (A)	R^2 (A)	RMSE (B)	R^2 (B)
Linear Regression	1.790	0.793	3.750	0.091
Ridge ($\alpha=1$)	1.790	0.793	3.749	0.091
Lasso ($\alpha=0.01$)	1.777	0.796	3.742	0.094
Decision Tree	1.950	0.754	4.209	-0.146
Random Forest	1.705	0.812	3.702	0.114
Gradient Boosting	1.677	0.818	3.687	0.121

Model	RMSE (A)	R ² (A)	RMSE (B)	R ² (B)
SVR (RBF)	2.002	0.741	3.397	0.254
Tuned Random Forest	1.669	0.820	—	—

Table 2: Regression Model Comparison — Scenario A (with G1/G2) vs Scenario B (without G1/G2)

4.3.3 Regression Diagnostics

Residual analysis of the tuned Random Forest (Scenario A) reveals a broadly normal residual distribution (Shapiro-Wilk $p = 0.0002$, indicating slight non-normality at the tails — expected given the integer nature of grades). The Residuals vs Fitted plot shows no systematic pattern, confirming that the model is not systematically biased across the prediction range. The learning curve shows the cross-validation RMSE stabilising at approximately 700 training samples, indicating that the current sample size is sufficient and further data collection would yield diminishing returns.

4.3.4 Feature Importance

Permutation importance analysis (15 repeats, ± 1 SD reported) for the tuned Random Forest (Scenario A) confirms G2 as the dominant predictor, followed by G1. Among non-grade features, failures, absences, and Medu are the three most important. This hierarchy is consistent with Cortez & Silva (2008), who report the same ordering across neural network and random forest models on the same dataset.

► Removing G1 and G2 increases RMSE by approximately 1.73 points — roughly equivalent to the average student improving by 1.73 grade points mid-year. The SVR model in Scenario B ($R^2 = 0.25$) demonstrates that behavioural and demographic features alone carry non-trivial predictive signal, making enrolment-time screening viable as a screening tool even without any prior academic data.

4.4 Results

RQ2: Which model best predicts final grade G3, and how much do prior grades contribute?

- Scenario A best model: Tuned Random Forest (RMSE = 1.669, $R^2 = 0.820$).
- Scenario B best model: SVR with RBF kernel (RMSE = 3.397, $R^2 = 0.254$).
- Removing G1/G2 increases RMSE by approximately 1.73 points, quantifying the information value of mid-year assessments.
- The most important non-grade features are: failures, absences, and Medu (mother's education).
- The learning curve confirms the model generalises well with no indication of heavy overfitting.

5. Classification: Predicting Pass or Fail

5.1 Substantive Issue

While regression predicts the continuous grade, a classification model that directly predicts pass ($G3 \geq 10$) or fail ($G3 < 10$) is more actionable for schools implementing a binary early-warning system. We address RQ3 by training multiple classifiers under both Scenarios A (with G1/G2) and B (without G1/G2), and by analysing the optimal decision threshold for deployment.

The dataset is imbalanced: 78.0% of students pass (814 Pass vs 230 Fail). A naive classifier that always predicts 'pass' achieves 78.0% accuracy without learning anything meaningful. Therefore, we evaluate models using F1-score, precision, recall, and AUC-ROC in addition to accuracy — with particular focus on recall for the Fail class, as missing an at-risk student (false negative) carries a higher social cost than offering support to a student who would have passed (false positive).

5.2 Methodology

Six classifiers were trained: Logistic Regression, Decision Tree (max depth=6), Random Forest (200 trees), Gradient Boosting (200 estimators), SVM (RBF kernel, probability=True), and K-Nearest Neighbours (k=7). Both scenarios used an 80/20 stratified train-test split. Linear and distance-based models (Logistic Regression, SVM, KNN) received standardised inputs; tree-based models used raw values. 5-fold cross-validation F1-score was used as the model selection criterion. The best model (Random Forest) was tuned via Stratified 5-fold GridSearchCV over the same hyperparameter grid as regression. Permutation importance and a threshold analysis (thresholds from 0.1 to 0.9) were additionally performed for Scenario B to guide deployment decisions.

For interpretability, a shallow Decision Tree (max depth=4) was also trained under Scenario B to produce human-readable classification rules suitable for use by non-technical school staff.

5.3 Analysis

5.3.1 Scenario A — With G1 and G2

All classifiers achieve accuracy above 89% in Scenario A, far exceeding the 78% naive baseline. The tuned Random Forest achieves accuracy = 0.900, AUC = 0.956, and F1 = 0.936. The Logistic Regression model performs comparably (AUC = 0.959), confirming that much of the discriminative signal from G1/G2 is linearly separable. The high AUC (> 0.95) indicates excellent discrimination between pass and fail students when prior grades are available.

5.3.2 Scenario B — Without G1 and G2

Removing G1 and G2 degrades all classifiers substantially. The tuned Random Forest achieves accuracy = 0.799, AUC = 0.663, and F1 = 0.881. Importantly, the recall for the Fail class drops to 0.24 at the default threshold of 0.5 — meaning 76% of failing students are missed. This is unacceptable for a real-world early warning system and motivates threshold analysis.

Model	Acc (A)	AUC (A)	F1 (A)	Acc (B)	AUC (B)	F1 (B)
Logistic Regression	0.914	0.959	0.945	0.775	0.671	0.867
Decision Tree	0.919	0.905	0.949	0.785	0.640	0.873
Random Forest	0.904	0.954	0.939	0.770	0.668	0.863
Gradient Boosting	0.900	0.942	0.936	0.766	0.654	0.859
SVM (RBF)	0.900	0.937	0.938	0.794	0.666	0.880
K-Nearest Neighbours	0.804	0.791	0.885	0.780	0.639	0.871

Model	Acc (A)	AUC (A)	F1 (A)	Acc (B)	AUC (B)	F1 (B)
Tuned Random Forest	0.900	0.956	0.936	0.799	0.663	0.881

Table 3: Classification Model Comparison — Scenario A (with G1/G2) vs Scenario B (without G1/G2)

5.3.3 Threshold Analysis

To address the low recall problem in Scenario B, we evaluated precision, recall, and F1 across 80 decision thresholds from 0.1 to 0.9. The F1-optimal threshold is 0.39 (compared to the default 0.50). Lowering the threshold from 0.5 to 0.39 increases recall for the Fail class from 0.24 to approximately 0.60, at the cost of reduced precision (more false alarms). For schools whose primary concern is detecting at-risk students, a threshold of 0.30-0.35 is recommended: this achieves recall ≈ 0.75 while precision remains above 0.40, meaning roughly 3 in 4 failing students would be flagged and only 60% of flagged students would turn out to have passed.

► The default threshold of 0.50 misses 76% of failing students in the deployment scenario. Lowering the threshold to 0.35-0.39 is strongly recommended, as the cost of a missed failing student (no support provided) is substantially higher than the cost of a false alarm (support offered to a student who would have passed anyway).

5.3.4 Interpretable Decision Tree

A depth-4 Decision Tree trained without G1/G2 achieves accuracy = 0.794 and produces fully human-readable classification rules. The root split is on failures (≤ 0.5 vs > 0.5): students with no prior failures have a 95%+ pass probability subject to absences and family support conditions. Students with prior failures face substantially higher fail risk, particularly those with higher absenteeism and fathers employed in services. This rule-based model is the most deployable option for school counsellors who need to explain decisions to parents and governors without technical ML expertise.

5.3.5 Cross-Subject Generalisation

A Logistic Regression model trained exclusively on Maths students and tested on Portuguese students (using only shared features, no grades) achieved accuracy = 0.735 and F1 = 0.833. The reverse direction (Portuguese $\rightarrow$ Maths) achieved accuracy = 0.709 and F1 = 0.818. These results confirm that behavioural and demographic predictors of pass/fail are substantially shared across subjects, supporting the deployment of a single school-wide model rather than separate subject-specific models.

5.4 Results

RQ3: Can we accurately identify at-risk students, and what threshold maximises detection?

- Scenario A (with G1/G2): Tuned Random Forest achieves AUC = 0.956, accuracy = 0.900. Excellent early-term identification tool once periodic grades are available.
- Scenario B (without G1/G2): Tuned Random Forest achieves AUC = 0.663, accuracy = 0.799. Viable enrolment-time screening, but threshold must be lowered.
- The F1-optimal threshold is 0.39; a threshold of 0.30-0.35 is recommended when recall of failing students is prioritised over precision.
- The interpretable depth-4 Decision Tree (accuracy = 0.794, no G1/G2) produces actionable rules for non-technical staff: the most important split is on past failures.
- Cross-subject validation confirms that the model generalises across Mathematics and Portuguese, justifying a unified school-wide system.

6. Discussion

6.1 Summary of Findings

This analysis demonstrates that student academic performance in secondary education is highly predictable using Machine Learning, and that the nature of the best-performing model and the magnitude of achievable accuracy are strongly dependent on whether mid-year periodic grades are available. When G1 and G2 are included, Random Forest achieves $R^2 = 0.82$ for grade prediction and $AUC = 0.96$ for pass/fail classification — performance comparable to the best models reported by Cortez & Silva (2008) on the same dataset. Without G1/G2, the best achievable AUC drops to 0.663, but the system remains practically useful, particularly when paired with an appropriate decision threshold.

The unsupervised analysis reveals that student behavioural profiles form two coherent groups strongly associated with academic outcomes, despite no grades being used in their formation. This finding aligns with sociological theories of educational attainment that emphasise family capital and lifestyle factors as foundational determinants of academic risk (Bourdieu, 1986).

6.2 Policy Implications

The analysis supports four concrete policy recommendations for school administrators:

- Deploy a two-stage early-warning system: at enrolment, use the Scenario B Random Forest with threshold 0.35 to generate an initial risk score from demographic and behavioural features. After the first grading period, update with G1 and G2 for a more accurate Scenario A prediction.
- Assign students in the K-Means Cluster 0 profile (lower parental education, higher past failures, higher alcohol consumption) to a monitoring programme from the first day of school — this cluster has a pass rate of only 66.8%.
- Prioritise reducing absences and supporting students with prior failures, as these are the top actionable predictors identified by permutation importance in both regression and classification tasks.
- Use the interpretable depth-4 Decision Tree to communicate risk decisions to parents and governors: its rules are transparent, auditable, and do not require technical expertise to understand or challenge.

6.3 Limitations

Several limitations should be acknowledged. The silhouette score for $K = 2$ clustering (0.118) is relatively low, suggesting that the student population does not separate into sharply distinct clusters — the two profiles overlap substantially. The Scenario B classification AUC of 0.663 is modest; schools should not use the enrolment-time model as the sole basis for high-stakes decisions. The dataset is from Portuguese schools in 2008 and may not generalise to other educational systems, countries, or time periods. Finally, the $G3 = 0$ records (5.1% of the sample) may represent withdrawal events that introduce confounding into the regression models.

7. Conclusion

This report has applied a comprehensive suite of Machine Learning techniques to the Student Performance dataset, addressing research questions across unsupervised learning, regression, and classification. The key finding is a clear trade-off between model deployability and predictive power: models trained with G1 and G2 achieve state-of-the-art performance ($R^2 = 0.82$, $AUC = 0.96$) but can only be used mid-year, while enrolment-time models achieve moderate performance ($AUC = 0.663$) but enable earlier intervention.

The most important student-level predictors across all three tasks are past failures, absenteeism, and parental education — all of which are known and partially modifiable at enrolment. The K-Means clustering identifies a meaningful at-risk student profile whose pass rate (66.8%) is 17.6 percentage points below the on-track profile, confirming that unsupervised methods can provide actionable early risk stratification without requiring any academic grades.

Collectively, these findings support the development of a two-stage, threshold-adjusted school early-warning system grounded in interpretable Machine Learning — one that is both technically sound and accessible to non-technical school staff and policymakers.

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